Enhancing interlayer bond strength of 3D-printed concrete through microstructural densification by means of incorporating sugarcane bagasse ash

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Abstract

3D-Printed Concrete (3DPC) is fabricated layer by layer, generating interlayer regions that typically exhibit elevated porosity and discontinuities. These interlayer interfaces act as mechanical weak links that govern failure initiation and load transfer across printed filaments, making interlayer bond strength a primary performance limiter. This study investigates a sustainable route to strengthen interlayer adhesion while reducing interface porosity by partially replacing cement with sugarcane bagasse ash (SBA). A comprehensive experimental program combined tensile and pull-off testing with microstructural characterization by scanning electron microscopy (SEM), X-ray diffraction (XRD), and micro-computed tomography (μ-CT). Five series of mixtures were produced with SBA contents increasing from 0 % to 20 % in 5 % increments. The mix with 15 % SBA delivered the best overall performance, with tensile strength and bond strength increases of 10.0 % and 31.4 %, respectively, after 90 days of curing. SEM revealed a refined interface microstructure; XRD indicated increased calcite formation in SBA-modified specimens. μ-CT results showed a 69 % reduction in porosity for the 15 % SBA mix compared with the control at 90 days, together with a more compact pore network. These findings demonstrate that SBA is an effective partial cement replacement for densifying interface regions and improving interlayer adhesion, thereby advancing sustainable practices in 3D concrete printing.

Introduction

With rapid urbanization, the construction sector has experienced a growing demand for buildings while simultaneously facing numerous challenges. To address these, the sector has increasingly adopted additive manufacturing technologies [1], [2]. Among these, 3D printing, an emerging construction method with substantial potential, enables the precise fabrication of diverse structural forms and complex geometries directly from 3D model data [3], [4]. Compared with conventional construction methods, 3D printing offers key advantages, including reduced construction cost and time, minimized material wastage, enhanced productivity, and greater design flexibility [5], [6], [7]. Unlike conventional concrete casting, 3D-Printed Concrete (3DPC) is extruded layer by layer to construct structural elements without the use of formworks, offering enhanced design flexibility and automation [8], [9], [10].

Although 3DPC exhibits significant advantages, the layered deposition inherent to the process introduces a critical performance challenge: the formation of weak and porous interlayer bonding zones. These interface regions are often characterized by higher porosity and lower strength compared to the bulk material, leading to pronounced mechanical anisotropy and compromised load-bearing capacity in printed elements [9], [10], [11]. Furthermore, the presence of numerous voids at the interface zone contributes to interlayer failure due to stress concentration and reduces the effective bonding area [12], [13]. Thus, interfaces play a critical role in 3DPC structures or elements, and research on layer-to-layer interface bond strength is steadily increasing. [11], [14], [15].

Existing research is primarily focused on three strategies for interlayer bond enhancement: (1) Process-based techniques [16], [17], [18], (2) Surface-based treatments [19], [20], [21], (3) Material-based approaches [22], [23]. Weng et al. [16] reported that interlayer bond strength in extrusion-based 3DPC is enhanced by higher superplasticizer dosage, increased pump speed, and water curing, with reduced thixotropy at the microscale being the most critical factor for interface quality. They identified surface moisture content and thixotropy as key factors governing interfacial bonding performance. Based on surface-based treatments, Lucen et al. [21] enhanced interlayer bonding in 3DPC by spraying a CO_2-activated interface enhancer during printing. The improvement was attributed to carbonation-induced formation of calcium carbonate and silica gel, which densified the interface and promoted mechanical interlocking. In certain structural conditions, adjustments to printing parameters or surface treatments are limited by equipment constraints, motivating material-based strategies to enhance interlayer performance [24]. Extending interlayer-focused studies to material aspects, Kaur et al. [22] evaluated the mechanical and durability performance of 3DPC incorporating fly ash and limestone–calcined clay binders. Their results showed that extrusion-induced densification and binder composition significantly reduced porosity and shrinkage, leading to improve strength and durability.

Although supplementary cementitious materials (SCMs) have been widely investigated to improve the sustainability and printability of 3D-printed concrete, existing studies have predominantly focused on bulk rheological performance, hydration behavior, and directional mechanical properties rather than interlayer-specific mechanisms. Zhang et al. [25] demonstrated that incorporating finely ground waste-derived ash into 3DPC enhances early-age structuration, promotes additional C–S–H formation, and reduces mechanical anisotropy through combined filler and pozzolanic effects. Consequently, the potential of alternative pozzolanic materials to refine the interface microstructure and thereby improve interlayer bond strength in 3DPC remains inadequately documented. In particular, sugarcane bagasse ash (SBA) — a reactive pozzolanic by-product of the sugar industry — has not been explored as a material-based approach to strengthen interlayer adhesion without reliance on surface treatments or adhesives. To address this gap, the present study investigates SBA as a partial cement replacement in 3DPC and examines its effects on interlayer bond strength, interface porosity, and curing-age performance, providing new evidence of SBA's efficacy for enhancing interlayer integrity in 3DPC.

The structure of the paper is as follows: Section 2 describes the experimental program and methodologies adopted for bond strength and microstructural evaluation. Section 3 presents the experimental results, including bond strength, crystalline phases, interfacial morphology, and pore structure. Section 4 provides an in-depth discussion, and Section 5 summarizes the main conclusions drawn from this study.

Materials

Ordinary Portland cement (P.I. 42.5) with a specific surface area of 340 kg/mm^2 and specific gravity of 3.16 was used. ISO Standard sand was used. A polycarboxylate-based superplasticizer (PCA-I) and hydroxyethyl methyl cellulose (HEMC, viscosity ≈100,000 mPa·s) were employed as the superplasticizer (SP) and viscosity-modifying agent (VMA), respectively. Raw SBA was sourced from the Guangxi Rural Sugar Industry (Nanning, China) and combusted at 800–1000 °C in a cogeneration boiler. The chemical

Tensile strength of intralayer concrete

The tensile strength development of concrete containing SBA was investigated at 3, 7, 14, 28, 56, and 90 days, as illustrated in Fig. 3. The tensile strength of cast specimens demonstrated a significant enhancement with increasing SBA content up to the M-15 mix, indicating that moderate SBA replacement effectively enhances matrix cohesion. At 28 days, the highest tensile strength was recorded at 4.91 MPa for M-15, reflecting a 7.1 % improvement over the CM.

Tensile strength continued to increase

Correlation between interlayer bond strength and intralayer tensile strength

A strong linear correlation was observed between the tensile strength of cast specimens and the bond strength of printed specimens across all mixes, shown in Fig. 9 with coefficients of determination (R^2) exceeding 0.86. This indicates that the intrinsic tensile capacity of the bulk mortar largely governs the interface bond behaviour in 3D-printed elements. As the tensile strength increases with curing age and improved matrix densification, a corresponding enhancement in bond strength is

Summary and conclusions

This study investigated the interlayer bonding strength of SBA-3DPC and characterized the pore structure and interface morphology through μCT and SEM. In addition, the influence of material parameters on interlayer performance was systematically examined. Based on these analyses, the following conclusions were established.

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Incorporation of SBA significantly improved both tensile and interlayer bond strength, with the 15 % replacement mix exhibiting the optimal performance. After 90 days, tensile

CRediT authorship contribution statement

Yong Yuan: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Syed Muhammad Mudassir Zia:** Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Taimur Mazhar Sheikh:** Software, Data curation. **Muhammad Irfan-ul-Hassan:** Writing – review & editing. **Xu-Peng Yao:** Supervision, Data curation. **Hong-Bo Zhu:** Data curation. **Jiao-Long Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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